P4 - Generative Architectural Envelope

ARC 3133-001 / Architectural Visualization Methods 1 / Fall 2026 - Full Term

Approach: Generative

Revision: 2026-10-19. Current assignment requirements and deadlines. Completed midterm work is reviewed using the evidence already submitted; no retrospective deliverables are added.

4.1 - Generative fields - a two-parameter matrix

Introduced: Week 10 - Oct 27 / **Due:** Week 11 - Nov 3, 11:59 PM Eastern Time / **Weight:** 10%

Purpose

Use the instructor-provided generative texture and SDF/volume groups to explore how two controls change a volume. Build the comparison matrix yourself.

What you will learn

- Explain the supplied groups through their inputs, outputs and visible effects.
- Use repetition to organize a systematic two-parameter comparison.
- Read the matrix, identify relationships and select promising spatial results.

What to do

- Choose two meaningful parameters, such as texture scale and threshold. Set five distinct values for each.
- Build a 5 x 5 matrix: one parameter changes across columns, the other down rows. Each of the 25 volumes represents one combination. The instructor supplies field tools, not a completed matrix.
- Use the same isometric view, material, lighting and display scale for all cells. Keep other design settings fixed. Label both axes and their five values.
- Mark promising results and briefly explain what each parameter changes and which result you want to develop. Keep the graph editable; do not manually model 25 unrelated objects.
- No separate point-cloud or site-analysis submission is required. You do not need to rebuild the internal SDF tools.

What to submit

- **Presentation:** one 17 x 11 inch landscape sheet, `Lastname_Firstname_P4a.pdf`, due November 3.
- **Matrix:** 25 isometric volumes arranged in five rows and five columns. Name the two parameters and label their five values along the corresponding axes; each cell must show the matching pair of settings.
- **Study notes:** use consistent view, lighting, material and display scale. Mark the result selected for 4.2 and add short observations explaining what the two parameters change and why that result is promising.
- **Editable source:** include your student-built matrix, the instructor-provided field groups with credit, and all texture/data dependencies. The graph must produce the 25 combinations and retain the selected settings.
- **Package:** `Lastname_Firstname_P4a/`. Match each row/column label to the working controls and check that the selected result can be reopened for 4.2.

What we assess

- A working student-built matrix, controlled variation of two parameters, readable comparison and thoughtful selection.
- Course weight 10%. Digital rubric: understanding 30%, communication 30%, technical execution 25%, exploration 15%.

4.2 - Architectural envelope - pavilion or small shop

Introduced: Week 11 - Nov 3 / **Due:** Week 12 - Nov 10, 11:59 PM Eastern Time / **Weight:** 15%

Purpose

Select a result from 4.1 and fine-tune its field into a compelling architectural envelope for a small pavilion or shop. Instructor-provided SDF Boolean operators are available to assist design development.

What you will learn

- Make architectural judgments about scale, enclosure, openings, occupied space and ground connection.
- Refine a generative result into one coherent proposal rather than collecting more unrelated variations.
- Understand how supplied SDF Boolean operations can combine or carve volumes to support architectural intentions.
- Use plan, section and isometric together to explain space and architectural intent.

What to do

- Choose a pavilion or small shop and state its intended use. Fit the entire envelope within 12 x 12 x 12 meters; these are maximum dimensions, not a required cube.
- Develop the selected field through its controls and necessary openings. Provide a clear entrance, usable occupied space, circulation and a deliberate relationship between ground, enclosure and roof.
- You may also use instructor-provided SDF Boolean operators - union, subtraction and intersection - to combine volumes, carve openings or refine the envelope. Use the operations that help your design; you do not need to use every operator or build the supplied tools yourself.
- Use openings to support access, light or views. Show scale with a person and a few key dimensions.
- Produce one plan, one section and one isometric from the same resolved model. Select drawing locations that explain the interior and its connection to the exterior.
- Focus on a convincing envelope and architectural visualization. Detailed services, full retail planning and construction documentation are outside this exercise. Keep the form suitable for contour fabrication.

What to submit

- **Presentation:** one 17 x 11 inch landscape sheet, `Lastname_Firstname_P4b.pdf`, due November 10.
- **Three drawings:** one plan, one section and one isometric of the same pavilion or small-shop envelope. Indicate the section cut on the plan; use clear drawing labels and scale information.
- **Architectural information:** state the intended use; identify the selected 4.1 result; show entrance/access, usable space, openings and the relationship to the ground. Include a person and key dimensions demonstrating the maximum 12 x 12 x 12 m envelope.
- **Design explanation:** briefly name the field controls and any supplied SDF Boolean operations used to develop the architecture. No additional Boolean comparison sheet is required.
- **Editable model:** retain the chosen field settings and any Boolean operations, with supplied groups credited and assets packaged in `Lastname_Firstname_P4b/`. Check that all three drawings come from this model and that it is ready for contouring.

What we assess

- Architectural clarity, quality of spatial refinement, consistent drawings and an understandable connection between the field controls, any Boolean operations used and design decisions.
- Course weight 15%. Digital rubric: understanding 30%, communication 30%, technical execution 25%, exploration 15%.

4.3 - Contour model - registration and assembly

Introduced: Week 12 - Nov 10 / **Due:** Week 13 - Nov 17, 11:59 PM Eastern Time / **Weight:** 10%

Purpose

Translate the same pavilion or shop envelope into contour parts and assemble a physical model that preserves its important spaces and openings.

What you will learn

- Relate model scale, material thickness and contour spacing.
- Use registration marks or alignment features to place parts accurately.
- Test, revise and document a fabrication and assembly strategy.

What to do

- Use the supplied contouring tools on the 4.2 envelope. Choose one slicing direction and spacing appropriate to the material and form; there is no fixed minimum contour count.
- Use 1:50 as the starting model scale: a 12 m envelope becomes 24 cm. Keep the completed model within the existing 30 cm limit on each axis; review another scale with your professor.
- Number the parts and include a registration mark or feature that establishes their alignment. Hold the model using glue, rods, tabs, spacers, a jig or another simple tested method.
- Test a small group of parts before the full cut. Check material thickness, kerf, alignment and how the assembly preserves openings and spatial qualities.
- Prepare clean full-size cutting curves and a layout within one lab-confirmed sheet of material. Confirm stock and machine access; optional printed components need their own allocation.

What to submit

- **Physical submission:** bring one assembled contour model of your 4.2 design to class on November 17. Use 1:50 as the starting scale and stay within 30 cm on each axis.
- **Presentation:** one 17 x 11 inch landscape fabrication sheet, `Lastname_Firstname_P4c.pdf`, due November 17 at 11:59 PM Eastern Time.
- **On the sheet:** chosen model scale, material/thickness, contour direction and spacing; a reduced numbered cutting layout; a registration/assembly diagram; short test-and-revision notes; one overall model photograph and one assembly/alignment detail.
- **Cutting file:** supply the full-size cutting curves separately in the FabLab-accepted format, with units and part numbers. The reduced layout on the PDF is not the machine file.
- **Supporting files:** editable model/contouring source, material and part list, and original photographs in `Lastname_Firstname_P4c/`. Include print/slicer files only if printed components were used.
- **Check:** part numbers agree across files and model, the registration method aligns the parts, the test informed the final cut, and the assembled model corresponds to the plan, section and isometric in 4.2.

What we assess

- Preparation and fabrication strategy 30%; fabrication quality 30%; visual documentation 25%; testing and revision 15%.
- Course weight 10%. Assess the parts, alignment, assembly and what the test improved, as well as fidelity to the architectural proposal.

Final booklet - submission checklist

Due: Final review - December 10-16, 2026; exact review slot to be confirmed / **Weight:** 10%

Reuse the assignment sheets and document selected revisions.

- Submit `Lastname_Firstname_BOOK.pdf` as a 17 x 11 inch landscape booklet. Include a cover, contents, all four projects in assignment order, references and brief captions identifying substantive revisions. There is no fixed total page count.
- Reuse the assignment sheets: 1 GSM; 2.1 process and 2.2 print documentation; 3.1 module; 3.2.1-3.2.3; 3.4 studies and 3.5 application; 4.1 matrix, 4.2 envelope and 4.3 fabrication. Include each shared sheet once. Completed midterm evidence remains valid.
- Include an overall image and useful detail of the CSG print and contour model, with short process/testing notes. Keep labels, typography and references consistent.
- Submit `Lastname_Firstname_GSM.pdf` separately with the revised four-page manual, plus the editable booklet layout and linked assets. Organize native project models and fabrication files in their assignment folders inside `Lastname_Firstname_BOOK/`.
- Open the PDF and source from the final package; check page order, missing links, readable images and drawing scales. Present the contour model and CSG documentation at the announced final review. The booklet deadline remains the December 10-16 final-review window; the exact slot will be announced.

Shared standards: Apply the GSM and credit sources. Package linked assets; check PDFs and editable files before uploading. Explain rules plainly and follow the course submission policies.